

Too Noisy at the Nanoscale?

—The Rise of Random Telegraph Noise (RTN) in Devices and Circuits

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Abstract—This paper gives an outline of our recent findings on the random telegraph noise (RTN) in nanoscale MOS devices and circuits.

Key words—Random telegraph noise (RTN); MOSFET

I. INTRODUCTION

The random telegraph noise (RTN), or random telegraph signal (RTS), in MOSFETs is not a new topic actually, which has been observed and studied in the 1980's [1]. However, recently it has attracted much more attention (e.g., [2-7]), due to its increasing amplitude as the device scaling down [2], which can hurt the circuit reliability due to the temporal variations (i.e., dynamic variability) caused by RTN [8]. Thus it is becoming “a designer's nightmare” rather than “a device physicist's dream” [9]. In this paper, we will give a short overview of our recent findings on RTN in nanoscale CMOS technology and circuits. (For other general and nice reviews, please refer to the classic [1] and recent ones [9-12].)

II. EXPERIMENTAL RESULTS

The RTN in a MOSFET originates from the stochastic trapping/de-trapping of the channel carrier into the switching oxide traps (or border traps) in the gate dielectrics [10], thus the capture/emission time constants are focused. Its characteristics is sensitive to technology (e.g., gate material and device structure). It is found that RTN is relatively faster (with smaller time constants) in high-k (HfO₂) devices than that in SiON (or SiO₂) devices. It is also found that in multi-gate device with ultra-thin body (e.g., FinFET and nanowire FET), the strong quantum confinement can enhance the bias dependence but alleviate the temperature dependence of RTN time constants [13-14]. From circuit application perspective, RTN is strongly dependent on device working conditions. It is found that RTN characteristics under AC large signal excitation (i.e., AC RTN) deviates from conventionally studied RTN under constant bias (i.e., DC RTN), that the time constants are different under DC and AC workloads with different frequencies [15-16].

III. MODELING AND IMPACTS ON DIGITAL CIRCUITS

Since it can lead to large V_{TH} (or I_D) degradation and cause transient probabilistic failure of digital circuits due to its stochastic nature, RTN will narrow the tight design margin in nano-CMOS technology. Therefore, it is very important to have compact modeling of AC RTN, for real circuit applications. Instead of trap occupancy probability under DC bias conditions

(p_{DC}), AC trap occupancy probability (p_{AC}) is proposed to investigate and model the dynamic trapping/de-trapping behavior of RTN [16]. Based on our newly proposed platform [17], the approach for accurate simulation of RTN effects in digital circuits is developed [18]. The impacts of RTN on digital circuits performances, e.g., jitters of ring oscillators (RO) and transient failure probabilities of SRAM are then evaluated [18-19]. The results show that degradations are highly workload dependent and that p_{AC} is critical in accurately evaluating the RTN induced performance degradation and variability.

IV. POTENTIAL SOLUTIONS AGAINST RTN

Generally, more guard-band, system redundancy, and error correction can be used to “ignore” RTN in digital design, but these solutions are becoming more and more costly and/or ineffective. We have proposed and demonstrated that, using the design with dynamic threshold MOS (DTMOS) is an effective solution of RTN suppression without performance penalty [20]. The results show that RTN amplitude is greatly reduced in DTMOS mode, which is even better than the body-biasing technique of FBB [21], due to the efficient dynamic modulation mechanism. Circuit stability and performance degradation induced by RTN are much improved in the design using DTMOS [20].

V. FROM SIMPLE RTN TO COMPLEX RTN

To extend the understanding of single-trap induced 2-state RTN (simple RTN), to more-traps induced multi-state RTN (complex RTN), in a single device, one usually assume that the traps in the device are independent from each other, thus conventionally complex RTN is treated as the superposition of individual traps independently (i.e., N traps lead to 2^N RTN states). This is good enough for the first-order estimation of RTN impacts on circuits, just including the spatial distributions of individual RTN parameters (amplitudes and time constants), and can also add the effects of spatial process variations and temporal transistor aging [17,19].

For some applications (e.g., low- V_{DD} or near- V_{TH} designs), however, the higher-order effects, especially the coupling between multi-traps in complex RTN cannot be neglected [22-24]. In special case studies, evident trap coupling is observed in DC/AC RTN amplitude and time constants [22], and the coupling strength exhibits nonmonotonic bias-dependence, which is due to the two competitive mechanisms of Coulomb repulsion and channel percolation conduction [23]. These trap coupling effects result in the

state-loss in complex RTN statistics (i.e., extreme-low probability to have 2^N RTN states) [24]. Two modes of statistical trap interactions are modeled [24]. The results indicate that, precise prediction of practical RTN impacts on circuit stability should be with the new understanding of trap coupling in complex RTN [22,24].

VI. SUMMARY AND OPEN QUESTIONS

The AC effect and trap coupling effect of RTN in nanoscale MOS devices and their impacts on digital circuits, as well as potential design solutions are investigated. Compact models and circuit simulation platform for RTN are developed. There are still some open questions remained for RTN, such as, understanding RTN amplitude from microscopic modeling (e.g., [25-29]), origin of the switching oxide traps (e.g., [30-31]), metastable defect states (e.g., [30-33]), connections between RTN and BTI (e.g., [10,12,33-34]), careful data analysis of RTN measurements (e.g., [10,11,35]), process dependence of RTN parameters (e.g., [36-37]), etc., which require further studies.

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